CptS 484: Software Requirements

WRS Evolution

Requirements Elicitation

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October 18, 2026

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Revision History

Date	Version	Changes	Editor
9/25/26	1.0	Created Initial Document with section 1	Jim
10/14/26	1.1	Created section 2 and 3	Jim
10/15/26	1.2	Worked on section 4 problems, goals, stakeholders, and functional requirements	Jim
10/17/26	1.3	Created section 5, 6, and 7	Jim
10/18/26	1.4	Worked on section 4 and 8	Santiago

[1] Introduction

1.1. Purpose

Indoor navigation presents severe accessibility challenges for visually impaired individuals. Navigating indoor spaces requires high-precision location determination, route selection, directional orientation, turn recognition, obstacle avoidance, and destination verification. Research on indoor navigation supports treating localization, movement direction, landmarks, and uncertainty as distinct requirements concerns[125].

Theia is a proposed smartphone application designed to address this problem by providing safe indoor navigation guidance through audio and haptic feedback[3]. Core system concerns include safety, accessibility, reliability, and route efficiency. The project will use the perceivable, operable, understandable, and robust principles of WCAG 2.2 as a design-review reference for mobile interaction[4]. The project also accounts for secondary stakeholders, including building administrators, sighted assistants and emergency personnel.

1.2. Scope

The main scope of the project is to be a smartphone application to allow blind people to safely navigate indoors. As such the application will primarily focus on being accessible for blind people to use and for use navigating indoors. The phone camera will be used to help detect potential obstacles that may be encountered indoors such as walls, doors, objects, stairs, etc. Moving objects are not included. However, navigating outdoors is not included in the scope of this project, so safety focus will be put on obstacles that may be encountered indoors rather than outdoors.

1.3. Objectives and Success Criteria

The application will be a smartphone app which will allow anyone visually impaired to safely navigate indoors. As such the application should:

1. Allow blind people to navigate the app.
2. Accept a destination and create a route from the current location to the destination.
3. Give detailed instructions to lead the user to the destination.
4. Detect potential obstacles in the way of the user and give them a way to avoid it.
5. Make emergency calls or messages if the user is lost or hurt.
6. Be fast and responsive so the user can get real time updates.
7. Be easy to use and learn for blind people.
8. Be able to choose the route taken for speed, comfort, etc.
9. Be customizable to meet every user's needs.
10. Be able to accommodate future updates and various new features.

1.4. Definitions, Acronyms, and Abbreviations

Term	Definition
WRS – World Requirements Specifications	Document connecting the problem domain, stakeholder needs and detailed requirements.
W – World	The environment context the system will inhabit
R – Requirement	The goals and features the system must accomplish
S - Specification	The specified way the system achieves the requirements
Domain	The final area or space the program will be deployed in
FR - Functional Requirement	Specific features and behaviors the final product should contain
NFR - Non-Functional Requirement	General quality and constraints the final product should meet

1.5. Overview

Section 2 of the document will go over the preliminary definitions for the domain, functional, and non-functional requirements that were provided. Section 3 will then go over any issues present in those requirements that will need to be updated and propose options to fix those issues. Section 4 will then go over the updated improved objectives as well as the W and RS of the document to see how each of the improved objectives help achieve them. It will also go over stakeholders of the project.

Section 5 will go over the preliminary prototype for the project, with section 6 showing the prototype mockups. Section 7 will be the initial user manual for how to use the application, with section 8 being the traceability matrix for the project. The references can be found in section 9, and any appendices will be included after that.

[2] Preliminary Definition

2.1. Preliminary Domain

PD_ID	Preliminary Domain Description
PD1	Indoor navigation app on mobile devices for the blind.

2.2. Preliminary Functional Requirements

P FR_ ID	Preliminary FR Description
PFR1	Accept destination location from the user.
PFR2	Route path to reach destination with potential options for the user.
PFR3	Tell user to walk a distance.
PFR4	Tell user to stop to turn.
PFR5	Detect objects and inform user how to avoid collision
PFR6	Place emergency calls and messages when detecting a fall or being lost.
PFR7	Decide next action based on user schedule or habits.

2.3. Preliminary Non-Functional Requirements

PNFR_ ID	Preliminary NFR Description
PNFR1	Safely navigate users to destination
PNFR2	Find the fastest route to destination
PNFR3	Find comfortable route to destination
PNFR4	Fully useable by blind users.
PNFR5	Be ubiquitous.
PNFR6	Be customizable by the user.
PNFR7	Be extensible to accommodate various variations.

[3] Issues with the Preliminary Definition Given

3.1. Domain Issues

Domain Issue ID	Domain Issue Description	
DI1	PD_ID	PD1: Indoor navigation app on mobile devices for the blind.
	1. Ambiguous on what counts as indoors, and what type of mobile device it is on.	
	Option 1	Be more specific
	Option 2	Be more general
	Choice	Option 1
	Rationale	Option 1 allows me to be more specific on what the domain means with less ambiguity.
Revised wording	App on smartphone devices to help the blind navigate in indoor spaces, including multiple floors, classrooms, offices, bathrooms, lounges, elevators, etc.	

3.2. Functional Requirements Issues

FR Issue ID	Description	
FRI1	PFR_ID	PFR1: Accept destination location from the user.
	How to accept destination?	
	Option 1	Voice command input.
	Option 2	Touchscreen UI.
	Choice	Option 1
	Rationale	Needs to be accessible by the blind so cannot rely on visual elements. Though could have Option 2 as a alternate option.
Satisfied by	FR1	

FR Issue ID	Description	
FRI2	PFR_ID	PFR2: Route path to reach destination with potential options for the user.
	How would the system know the layout of an indoor space. How would the system know the current location of the user.	
	Option 1	Use wireless beacons and tools like Google Indoor Maps and Microsoft Soundscape.
	Option 2	Use phone camera to navigate buildings and read signs/maps.

	Choice	Option 1
	Rationale	Much easier and simpler, though could do option 2 if option 1 is not available.
Satisfied by	FR2	

FR Issue ID	**Description**	
FRI3	PFR_ID	PFR3: Tell user to walk a distance.
	How to inform the user.	
	Option 1	Text to speech voice
	Option 2	On screen interface with instructions
	Choice	Option 1
	Rationale	Needs to be useable by the blind. Can be supplemented with option 2.
Satisfied by	FR3	

FR Issue ID	**Description**	
FRI4	PFR_ID	PFR4: Tell user to stop to turn.
	How to detect when the user is at the right spot to stop and turn.	
	Option 1	Use phone camera to detect junction to turn.
	Option 2	Use Microsoft Soundscape.
	Choice	Option 2
	Rationale	Rely on advanced sensors rather than camera detection, though could use option 1 as backup.
Satisfied by	FR4	

FR Issue ID	**Description**	
FRI5	PFR_ID	PFR5: Detect objects and inform user how to avoid collision
	How to detect the object.	
	Option 1	Use phone camera with object recognition.
	Option 2	Use external tools.
	Choice	Option 1
	Rationale	Better to be all in one for the app, and be able to recognize it with the camera.
Satisfied by	FR5	

FR Issue ID	**Description**

FRI6	PFR_ID	PFR6: Place emergency calls and messages when detecting a fall or being lost.
	How to know when the situation is an emergency	
	Option 1	Use haptic sensors to detect fall and navigation/timer to detect if the user is lost
	Option 2	Use voice command to let user inform app that it is an emergency
	Choice	Option 1
	Rationale	User may not be able to speak in an emergency, though could also include option 2
Satisfied by	FR6	

FR Issue ID	**Description**	
FRI7	PFR_ID	PFR7: Decide next action based on user schedule or habits.
	How to predict?	
	Option 1	Use the current time to keep track against previous schedules/habits
	Option 2	Memorize common routes the user takes.
	Choice	Option 1
	Rationale	Able to predict and automatically prepare common actions
Satisfied by	FR7	

3.3.Non-Functional Requirements(NFR) Issues

NFR Issues ID	**Description**	
NFRI1	PNFR_ID	PNFR1: Safely navigate users to destination
	How exactly is safe defined	
	Option1	User did not collide with anything
	Option2	User did not get any injury/minor scrape
	Choice	Option 1
	Rationale	Best for the user to not collide with any obstacle or trip on anything.
Satisfied by	NFR1	

NFR Issues ID	**Description**	
NFRI2	PNFR_ID	PNFR2: Find the fastest route to destination
	What does it mean to be fast?	

	Option1	No other route would be faster
	Option2	Fast within reason, without going through any unsafe or restricted area
	Choice	Option 2
	Rationale	Needs to be fast, but also be within reason and not endanger user.
Satisfied by	NFR2	

NFR Issues ID	**Description**	
NFRI3	PNFR_ID	PNFR3: Find comfortable route to destination
	What does it mean to be a comfortable route?	
	Option1	No major challenges such as unnecessary stairs when an elevator is available or crowded area.
	Option2	Route with minimal obstacles.
	Choice	Option 1
	Rationale	Allow multiple routes to be presented, with less stairs or minimal obstacles or less walking, etc.
Satisfied by	NFR3	

NFR Issues ID	**Description**	
NFRI4	PNFR_ID	PNFR4: Fully useable by blind users.
	What is usable by blind people and what is not.	
	Option1	App completely voice controlled and assisted
	Option2	Include some basic touch screen elements.
	Choice	Option 1
	Rationale	Primarily option 1 but can also add option 2 to help ease use for those with impaired vision
Satisfied by	NFR4	

NFR Issues ID	**Description**	
NFRI5	PNFR_ID	PNFR5: Be ubiquitous.
	What does it mean to be ubiquitous?	
	Option1	Keep track of any potential obstacles.

	Option2	Be available and useable by many different people.
	Choice	Option 2
	Rationale	To be useful to as many people as possible.
Satisfied by	NFR5	

NFR Issues ID	**Description**	
NFRI6	PNFR_ID	PNFR6: Be customizable by the user.
	What options are included.	
	Option1	Volume, instruction intervals, routes, etc
	Option2	Everything can be altered
	Choice	Option 1
	Rationale	Easier to implement and less likely to break anything.
Satisfied by	NFR6	

NFR Issues ID	**Description**	
NFRI7	PNFR_ID	PNFR7: Be extensible to accommodate various variations.
	How extensible should it be?	
	Option1	Allow different language options, new updates and features, upgraded hardware for future proofing.
	Option2	Allow updates.
	Choice	Option 1
	Rationale	Able to be iterated and updated in the future easier.
Satisfied by	NFR7	

[4] WRS

4.1.W

4.1.1. Problem

Problem ID	Problem Description	Corresponding Goals
P1	Blind people cannot navigate and create a route indoors.	G1, G2
P2	Blind people may not be able to avoid obstacles.	G3
P3	Blind people do not know when they reach their destination.	G1, G9
P4	Blind people do not know how long to walk and when to turn.	G1, G9
P5	Blind people may not be able to receive help if they get incapacitated or lost.	G7
P6	Blind people cannot navigate mobile apps.	G2, G9
P7	Users may not be able to use the app (compatibility).	G6
P8	Users should be able to customize options.	G4, G8
P9	Blind people cannot see anything on the screen.	G2, G9

4.1.2. Goals

Goal ID	Goal Description	Backward Traceability	Forward Traceability
G1	Create a fast route to get from current location to destination.	P1, P3, P4	IFRO2, IFRO3, IFRO4
G2	The app is fully operatable by blind people.	P1, P6, P9	IFRO1, INFRO4
G3	Detect obstacles and inform user how to avoid it.	P2	IFRO5, INFRO1
G4	Give options on what route to use.	P8	IFRO7
G5	Create a comfortable route to get from current location to destination.	P1	INFRO3
G6	App is available on common mobile devices.	P7	INFRO5
G7	App is updated and maintained.	P5	IFRO6
G8	App is customizable to suit the user's needs.	P8	INFRO6
G9	App is able to speak to user through text to speech (TTS) voice.	P3, P4, P6, P9	INFRO4

4.1.3. Improved Understanding of Domain, Stakeholders, Functional, and Non-Functional Objectives

4.1.3.1. Improved Domain

Improved Domain ID	Improved Domain Description
ID1	App on smartphone devices to help the blind navigate in indoor spaces, including multiple floors, classrooms, offices, bathrooms, lounges, elevators, etc.

4.1.3.2. Stakeholders

- Sponsors
  - Faculty: Bolong Zeng
- Potential Users of THEIA
  - Blind people who need a way of safely navigating indoors on their own.
  - Visually impaired people who need additional assistance navigating indoors
  - People who can be quickly alerted if anything happens to the user

4.1.3.3. Improved Functional Objectives

Based on the above information and our goals, the functional objectives of THEIA are:

Improved FR Objective ID	Objective Description	Alleviates Problems	Achieves Goals
IFRO1	Destination must be able to be input by the blind.	P1, P6	G2
IFRO2	The app must be able to know the building layout to create a route.	P1	G1
IFRO3	The app must be able to inform the user how far to travel.	P4	G1
IFRO4	The app must be able to keep track of user location.	P1, P4	G1, G5
IFRO5	The user must be able to be navigated safely.	P2, P3	G3
IFRO6	The user must be able to get help when they need it.	P5	G7
IFRO7	Users can get suggested routes without having to manually repeat them.	P7, P8	G4, G8

4.1.3.4. Improved Non-Functional Objectives

Improved NFR Objective ID	Objective Description	Alleviates Problem	Achieves Goal
INFRO1	Navigate the user safely without collisions.	P2	G3
INFRO2	Calculate the fastest safe route.	P1	G1
INFRO3	Offer comfortable routes (avoiding stairs).	P1	G5
INFRO4	Be fully usable by blind users (voice-first).	P6, P9	G2, G9
INFRO5	Be available on common mobile devices.	P7	G6
INFRO6	Be customizable (volume, intervals).	P8	G8
INFRO7	Be extensible for future updates.	P7	G7

4.2.RS

4.2.1. Functional Requirements

FR ID	Description
FR1	Accept destination location from the user through voice command.
Satisfies Functional Requirement Issue	FRI1
Satisfies Objectives	IFRO1
Satisfied by prototype feature	Voice Command / Home Screen

FR ID	Description
FR2	Route multiple paths to reach the destination using Indoor Maps and let the user select one to follow.
Satisfies Functional Requirement Issue	FRI2
Satisfies Objectives	IFRO2
Satisfied by prototype feature	Route Confirmation Screen

FR ID	Description
FR3	Guide the user to walk a certain number of steps using TTS voice.
Satisfies Functional Requirement Issue	FRI3
Satisfies Objectives	IFRO3
Satisfied by prototype feature	Navigation Screen

FR ID	Description
FR4	Using Soundscape and the phone camera to keep track of user location, tell user using TTS voice to stop at a certain point to turn.
Satisfies Functional Requirement Issue	FRI4
Satisfies Objectives	IFRO4
Satisfied by prototype feature	Navigation Screen

FR ID	Description
FR5	Use phone camera to detect potential obstacles to warn users using TTS voice of the danger and inform them how to avoid it.
Satisfies Functional Requirement Issue	FRI5
Satisfies Objectives	IFRO5
Satisfied by prototype feature	Hazard Detected Screen

FR ID	Description
FR6	Contact emergency contact when a fall is detected using phone sensors or when the app detects the user is lost.
Satisfies Functional Requirement Issue	FRI6
Satisfies Objectives	IFRO6
Satisfied by prototype feature	Emergency Alert Screen

FR ID	Description
FR7	Keep track of common user habits and input schedules to automatically prepare routes for the user to use.
Satisfies Functional Requirement Issue	FRI7
Satisfies Objectives	IFRO7
Satisfied by prototype feature	Home Screen

4.2.2. Non-Functional Requirements

NFR ID	Description
NFR1	The app must ensure the user does not collide with any static obstacles (walls, doors, chairs) during navigation.
Operationalized Functional Requirements	FR5
Satisfies Nonfunctional Requirement Issue	NFRI1
Satisfies Non-functional Objective	INFRO1
Constrains	Requires camera API access and <500ms processing latency.
Satisfied by prototype feature	Hazard Detected Screen

NFR ID	Description
NFR2	The app must calculate routes within 3 seconds of destination input.
Operationalized Functional Requirements	FR2
Satisfies Nonfunctional Requirement Issue	NFRI2
Satisfies Non-functional Objective	INFRO2
Constrains	Requires efficient algorithm and cached building maps.
Satisfied by prototype feature	Route Confirmation Screen

NFR ID	Description
NFR3	The app must offer at least one "comfortable" route that avoids stairs if an elevator is available.
Operationalized Functional Requirements	FR2
Satisfies Nonfunctional Requirement Issue	NFRI3
Satisfies Non-functional Objective	INFRO3
Constrains	Requires building layout data to tag stairs vs. elevators.
Satisfied by prototype feature	Route Confirmation Screen

NFR ID	Description

NFR4	The app must be 100% operable via voice commands and haptic feedback without requiring visual screen interaction.
Operationalized Functional Requirements	FR1
Satisfies Nonfunctional Requirement Issue	NFRI4
Satisfies Non-functional Objective	INFRO4
Constrains	Must utilize TTS (Text-to-Speech) and STT (Speech-to-Text) APIs.
Satisfied by prototype feature	All Screens

NFR ID	**Description**
NFR5	The app must be available on both iOS and Android platforms.
Operationalized Functional Requirements	All
Satisfies Nonfunctional Requirement Issue	NFRI5
Satisfies Non-functional Objective	INFRO5
Constrains	Development must use cross-platform frameworks or dual native development.
Satisfied by prototype feature	N/A (Architectural constraint)

NFR ID	**Description**
NFR6	The user must be able to change the TTS voice volume and the frequency of instructions.
Operationalized Functional Requirements	Options menu
Satisfies Nonfunctional Requirement Issue	NFRI6
Satisfies Non-functional Objective	INFRO6
Constrains	Must save user preferences locally.
Satisfied by prototype feature	Options Screen

NFR ID	**Description**
NFR7	The app must support multiple languages and allow for new features to be added in future updates.

Operationalized Functional Requirements	Options menu (Language)
Satisfies Nonfunctional Requirement Issue	NFRI7
Satisfies Non-functional Objective	INFRO7
Constrains	Code must be modular; TTS/STT must support localization.
Satisfied by prototype feature	Options Screen

4.2.3. Specifications

Functional Specification ID	Functional Requirement
FS1	The user can say "Create a route to [Room Y]" to start navigation.
Satisfies Functional Requirement	FR1
Satisfies Objectives	IFRO1, INFRO4
Satisfied by prototype feature	Voice Command Menu

Functional Specification ID	Functional Requirement
FS2	The app will provide audio options: "Fastest route (8 mins)" or "Comfortable route (12 mins)".
Satisfies Functional Requirement	FR2
Satisfies Objectives	IFRO2, INFRO3
Satisfied by prototype feature	Route Confirmation Screen

Functional Specification ID	Functional Requirement
FS3	The app will say "Walk 45 feet ahead" and "Turn left in 10 feet".
Satisfies Functional Requirement	FR3
Satisfies Objectives	IFRO3
Satisfied by prototype feature	Navigation Screen

Functional Specification ID	Functional Requirement
FS4	The app will track location via beacons/Soundscape and announce when to turn.

Satisfies Functional Requirement	FR4
Satisfies Objectives	IFRO4
Satisfied by prototype feature	Navigation Screen

Functional Specification ID	Functional Requirement
FS5	The app will say "Obstacle ahead, stop" and provide a detour instruction.
Satisfies Functional Requirement	FR5
Satisfies Objectives	IFRO5, INFRO1
Satisfied by prototype feature	Hazard Detected Screen

Functional Specification ID	Functional Requirement
FS6	If a collision is detected by the IMU, the app sends an emergency message to the designated contact.
Satisfies Functional Requirement	FR6
Satisfies Objectives	IFRO6
Satisfied by prototype feature	Emergency Alert Screen

Functional Specification ID	Functional Requirement
FS7	The app will prompt: "Would you like to use your usual route to the Cafeteria?"
Satisfies Functional Requirement	FR7
Satisfies Objectives	IFRO7
Satisfied by prototype feature	Home Screen

[5] Preliminary Prototype

5.1. Routing Component Description

Routing consists of the steps the user and the application takes to create a route the user can use to navigate to their destination.

Input

The input for the Routing component will be the user telling the app where their desired destination is. The user can input the destination by using a voice command anywhere in the app. Wireless beacons set up in the buildings can be received by the app to determine the current room. Start location can also be input manually if the app cannot automatically detect it.

Output

The output will be a list of routes the app has computed, allowing the user to select one to start navigation. The list of routes will be told to the user using a text-to-speech (TTS) voice where the user can select one using their voice. The list of routes can also be displayed and selected on the screen for users who can see.

Implementation

The application will create routes using the A-Star algorithm. Using the detected building layout from wireless beacons and Google Indoor Maps, the start location detected by wireless beacons, and the destination, the application can use the algorithm to find the path with the shortest distance. The program can also add weight to a path that has documented obstacles to calculate a more comfortable and safer route. Once these routes have been created, it can output them for the user to decide.

Validation

- Algorithm Validation: Test various cases, such as routes with known fastest or most comfortable routes or even impossible routes to ensure output is accurate.
- Layout Graph Network Validation: Run scripts to check the graph network for the layout of the building is valid, having no logical errors to ensure it can be relied on to create a route.

5.2. Navigation Component Description

Navigation is the process of guiding the user to the destination safely using the selected route.

Input

The input for navigation will be the previously selected route along with the phone's camera and haptic sensors to detect any potential obstacles.

Output

The output will be a TTS voice to guide the user to their destination, along with warnings about potential obstacles and instructions to avoid them. If the app detects the user is incapacitated or lost, it can also send an emergency message.

Implementation

Using APIs such as Google ARCore Depth and Google ML Kit for Android, or Apple ARKit and Apple CoreML for iOS devices, the phone camera can be used to scan where the user is in space and detect objects. Using Microsoft Soundscape, the app can detect any walls or obstacles as well, guiding the user to their destination. If the app detects an object or other obstacle in the user's path, it can warn the user and scan the area to tell the user how to avoid it. As the user approaches the destination, the app can detect they have arrived and end navigation.

The phone's Inertial Measurement Unit (IMU) can be accessed to detect sudden collisions, while being lost can be measured if the current location does not seem to match the app's route and building layout. In these cases, an emergency message can be sent out either automatically by the app or at the user's request through a voice command.

Validation

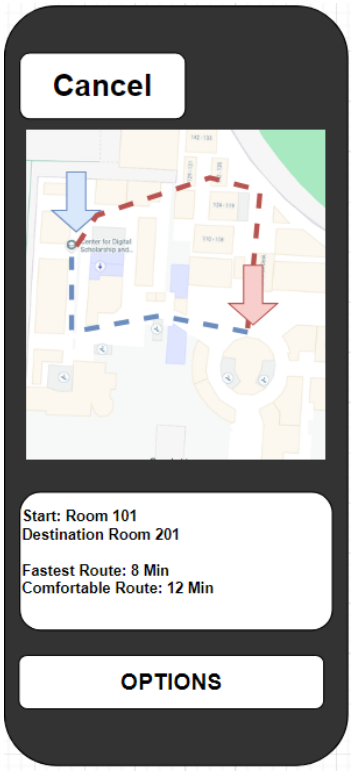
- Environmental Calibration: Ensure the app can successfully navigate and detect objects in various conditions such as low-light settings, and high glare settings, as well as not get confused by things such as mirrors and glass.
- Obstacle Detection Validation: Ensure the app can accurately measure space and distance of objects and walk next to objects and walls without getting confused.
- Drop Detection Calibration: Ensure the app is properly calibrated, so a drop is detected without fail, but also is not triggered where there is no collision.

[6] Prototype Interface Mock-ups

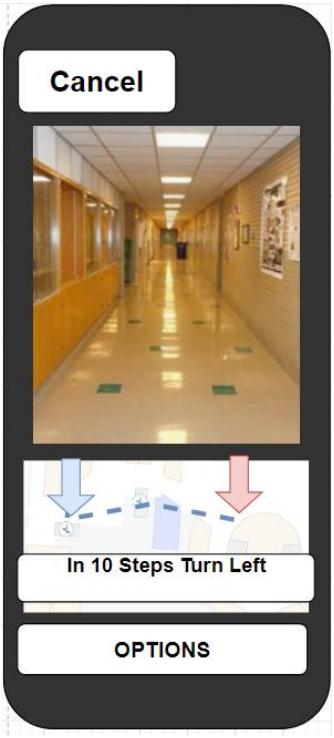
6.1. Scenario 1: Navigation



Bob is currently in room 101 and wants to go to room 201. He opens the app and the app starts talking to Bob giving instructions on how to use the app and his options such as starting navigation or going to options. Bob using the voice commands states “Create a route to room 201.” The app then switches to the map screen.



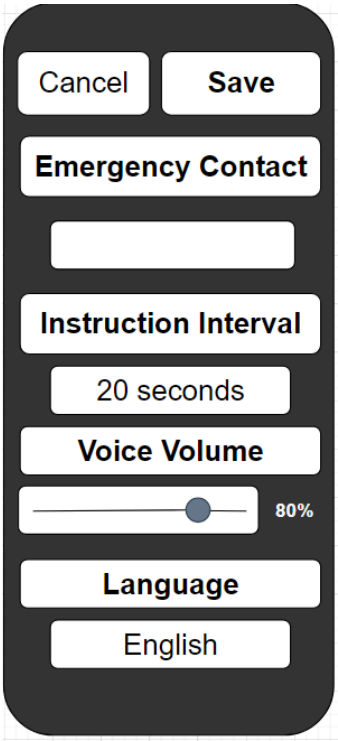
The app calculates several routes and presents those options to Bob out loud. “The fastest route will take 8 minutes; the most comfortable route will take 12 minutes.” Bob then states he wants to use the fastest route. The app starts navigation using that route and instructs Bob to point the camera forward as it starts giving instructions.



Using the instructions Bob navigates the route, with the app detecting obstacles and warning Bob so he can avoid it. Once he reaches the destination the app tells Bob he has reached the destination.

6.2. Scenario 2: Options

Bob wants to customize some things about the app so he tells the app “Go to options.” And the app opens the options screen.



The app starts listing out the options Bob can change and what they currently are set to, such as the emergency contact, instruction interval, voice volume, and language. Bob states he wants to change the instruction interval to 40 seconds. The app changes the instruction interval, then Bob says to go back. The app confirms with Bob if he wants to save the latest changes before going out of the options menu and back to the previous screen.

[7] User Manual

Introduction

THEIA is a navigation app designed for blind or visually impaired people to safely navigate indoors to their destination. The app allows the use of voice commands to navigate the app so looking at the screen is not required for use.

Menu

On the main menu screen the user can select to start navigation, show a list of saved routes, and go to options.

Voice Commands

- Start Navigation: Go to the Navigation screen.
- Create a Route to Y: Go to the Navigation screen and create a route from the current location to Y.
- Create a Route from X to Y: Go to the Navigation screen and create a route from the current location to Y.
- Saved Routes: Go to list of saved routes.
- Use saved route from X to Y: Start navigation using the saved route from X to Y.
- Options: Go to options menu.

Options

In options various options can be customized. Emergency contacts is the list of contacts to be contacted if a collision is detected or the user is lost. By default, 911 is included. Instruction interval specifies how long there is in between each instruction for navigation. Voice volume is the volume of the speaker TTS voice of the app. Language specifies the app's language.

Voice Commands

- Go Back/Return: Confirms whether user wants to save changes before returning to previous screen.
- Cancel: Return to previous screen.
- Save: Saves the changes then return to previous screen.
- Add Emergency Contact X: Add X to list of contacts to be contacted if the user needs help.
- Remove Emergency Contact X: Removes X from list of contacts.
- Change Instruction Interval to X: Changes instruction interval to X.

- Change Voice Volume to X: Changes voice volume to X.
- Lower/Raise Voice Volume by X: Changes voice volume by X. is no number was specified change it by 10%. Warn user if voice will be set to 0%.
- Change Language to X: Change language to X.

Saved Routes

In Saved Routes the user can view a list of saved routes and can choose one to repeat the same route. They can also choose to delete a saved route.

Voice Commands

- Use route X to Y/Use route #X: Starts navigation using the selected route. If there are multiple of the same routes from X to Y app asks for specification on which route to use.
- Delete route X to Y/Delete route #X: Deletes the saved route from X to Y or route #X. If there are multiple of the same routes from X to Y app asks for specification on which route to use.
- Go back/Return: Returns to the main menu screen.
- Options: Go to Options Screen.

Create Route

The user can create a new route in the Navigation screen by specifying their destination. The app will automatically create several routes to the destination, and the user can select one to use.

Voice Commands

- Create a Route to Y: Create a route from the current location to Y.
- Create a Route from X to Y: Create a route from the current location to Y.
- Saved Routes: Go to list of saved routes.
- Use saved route from X to Y: Start navigation using the saved route from X to Y.
- Options: Go to options menu.
- Go back/Return: Return to previous screen.
- Use fasted route: Start navigation using the fastest route.
- Use most comfortable route: Start navigation using the most comfortable route.
- Use route #X: Start navigation using route #X.

Navigation

The app will instruct the user on how to reach their destination using step by step instructions, as well as instruct the user to hold their phone forward to use the camera. The

app will warn the user of any obstacles in their way and how to avoid it. Once the destination is reached the user can choose to save the route before the app returns to the main menu screen. If a collision is detected and the user does not respond the app will automatically send a message to an emergency contact.

Voice Commands

- Stop/cancel/return/go back: Stops navigation and returns to the list of routes or previous screen.
- How long until destination: Tells user how much longer it is to reach the destination.
- Where am I: Tells user what room they are currently in.
- Help/contact help/I am lost: The app will ask the user to select a contact to send a emergency message to.
- Options: Go to options menu.

[8] Traceability

Problem	Goal	Objective	Requirement	Specification
P1: Blind people cannot navigate/create routes indoors.	**G1:** Create a fast route.	**IFRO1, IFRO2, IFRO4**	**FR1, FR2, FR4**	**FS1, FS2, FS3, FS4**
P2: Blind people may not avoid obstacles.	**G3:** Detect obstacles and inform user.	**IFRO5, INFRO1**	**FR5, NFR1**	**FS5**
P3: Blind people do not know when they reach destination.	**G1, G9:** Create a fast route, speak to user.	**IFRO3**	**FR3**	**FS3**
P4: Blind people do not know when to turn.	**G1, G9:** Create a fast route, speak to user.	**IFRO4**	**FR4**	**FS4**
P5: Blind people may not get help if incapacitated.	**G7:** App is updated/maintained.	**IFRO6**	**FR6**	**FS6**
P6: Blind people cannot navigate mobile apps.	**G2:** App is fully operable.	**IFRO1, INFRO4**	**NFR4**	**FS1**
P7: Users may not be able to use the app.	**G6:** App is available on mobile.	**INFRO5, INFRO7**	**NFR5, NFR7**	N/A (Architectural)
P8: Users should be able to customize options.	**G8:** App is customizable.	**IFRO7, INFRO6**	**NFR6**	Options Screen
P9: Blind people cannot see the screen.	**G2, G9:** App speaks via TTS.	**INFRO4**	**NFR4**	All Screens

[9] References

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